

**HR Analytics**  
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**Week: 1**  
**Lecture 03: Data science framework**

Dear participant, in first two sessions, we have learned the basics of the HR analytics. So, in the first session, if you remember, we discussed the what are various names of HR right and how we should apply the basic concepts related to HR. So, that is the basic understanding that we had developed related to the HR analytics in first session. In the second session, we developed a basic understanding of the data and how we should develop and use the good data in order to analyse and make some results related to the data analytics. So, in addition to that, in first two sessions, in this session, we will learn a framework through which you can analyze the HR data. So this framework that you can say data science framework.

So any kind of analytics in any area that you are planning to apply this analytics, you can use this framework, right? So in HR area, if we are applying, so steps are same. In marketing, if you are applying or you are applying any other area, the steps are same. But in, so this framework that we will understand in HR context, right? So what are the steps? So here I have written the steps, all steps of that data science framework, right? So that is the content. So first step is to define the goal, right? So that we are keep discussing in our sessions.

If you have to start this analytics, first thing that you have to do, you have to define the goal, right? So as we know this is a course related to HR analytics, so that is why we have written. So, any HR problem is what you have to define. Second step is related to the definition of HR problem should cover. So, whatever problem that you are defining, so you have to clearly define the all things related to that HR problem. So, that is what you have to do.

Next step that you have to you have the moment you have defined the problem after defining it you have to collect the data related to it and manage it. So, manage it means data analysis, structured data, unstructured data, missing data, quality of data. So, these are the all issues that we had already discussed in the second, so second session. So, this is the second step of the data science framework, right. Third, after collecting the data you have to build a model, right.

So, in detail, we will discuss in depth how you can build a model in HR. So, basically, it is a matrix right that is a matrix that we need to develop. So, in the second session if you remember we had discussed the how to develop the matrix and what are the criteria for a good HR matrix right. So, here you have to build a model so that matrix that you have to develop through which you will analyze the data. The moment you have developed this model right then you have to evaluate and critique the model.

Why it is important? Because we should know all matrix that we are developing, we should ensure that they are able to solve the problems. Because all matrix that we develop related to our problem, they all are proxy, no matrix will give you the exact thing. But some matrix are better than others because they give you the closer they have proximity to the defined goal. So, in detail along with the example this thing also we will discuss. Next is the moment you have developed the model then you evaluate and critique the model.

You evaluate whatever what are the limitations and strengths of that particular matrix and whether it is able to capture the real phenomena or not. Where whatever that data wants to communicate whether you are able to understand or not through that matrix. So, these all things that you have to do in this particular step. So, in detail, we will discuss it, and then finally, you have to present the results related to that particular model, whichever model you have developed. If you feel yes this results are good and giving you the better understanding about the problem then you can deploy that particular model in your organization.

So, these are the various steps of the data science. So, whenever you are applying this HR problem, so I always suggest you go through this, these steps. If you will randomly you will do before, if you have not defined the problem and collecting the data and then presenting the result without evaluating that particular model that you have developed or that matrix. So, you may commit many mistakes while doing this process. So, I suggest all HR managers to follow these steps and apply the HR analytics tool.

So, let us start with first step defining the HR problem. So, being very clear on this is very important, and defining it is very important because the specific choices we make further about data and tools are fundamentally dependent on this. So, I hope you can understand why it is important. There are two reasons; the first reason is that if you are not clear about your goal and what exactly you want to do, then you will not be clear about the data, which data you are supposed to collect and which tool you are supposed to use. So, that is why it is important to define a problem.

So, how you can define a problem? Like, attrition, how many are leaving? So, then you can tell this is the percentage of people who are leaving from this particular department, this particular region people are leaving. right. So, you can do in-depth analysis of the attrition and then you will understand you have to collect the data related to the attrition, how many people are leaving the organization. Similarly, selection. So, what kind of people if you want to do the analysis of selected people.

So, out of 10,000 people let us assume you have selected 200 people. Now you want to do this analysis of the selected people, that quality of hire, right, how good these people are. So now you understand you have to collect the data related to the quality of hire because you want to understand how good they are. If you want to understand their demography then you will collect the data related to the demography. So if your problem statement is to evaluate the quality of people whoever, whoever have been selected.

So, you will collect the data related to the quality of hire, right? And your problem statement to understand the demography and demographics of the selected people from where they are, what is their age, what is their expectations, what they like, and what they do not like. If you want to understand these all things, then you will collect the data related to the demographic. So, that is what I was saying in the first step. It is very, very important to define the problem clearly. If you are unable to define the problem clearly you will not be able to select which type of data that you are supposed to collect and which tool that you need to apply to analyze that particular data.

So, that is why it becomes very very important to define the problem clearly, whether it is related to the any function. So, I already said to define this problem you can select a HR function, whether it is related to recruitment, selection, training, development, performance, compensation and then you can go in depth. Then you can go in-depth and understand what the problem is, and then you can decide which data you have to collect related to the recruitment and which tool you are supposed to apply. So, in the first session if you remember we had discussed the various types of the statistics. So, descriptive statistics, diagnostic statistics, predictive statistics and prescriptive statistics.

So, again which type of statistics that you will select that the method to analyze the tool related to which type of analytics that you will select, which tool that you will select related to which type of analytics tool that you will select. That also will depend on the goal. If your problem is related in term of exploring something, then you may select the tool related to the descriptive. Like for example, if you want to understand how many people are leaving, then here is a simple count that you can apply from which department they are leaving, which area they are leaving, and which gender is leaving more. So, problem is to understand that particular phenomena, that attrition phenomena.

So, simply that you can apply descriptive analytics tool. If your problem is, is there any relationship between age and people who are leaving? In that case you can apply this diagnostics. So, in that case you can plot a correlation graph. So, if you want to understand the relationship between work experience and the people who are leaving from the organization. So, if you want to understand who is leaving more, people who are having more experience or less experience, who are leaving more, in whom there is a high level of attrition rate, early joining people or people who are having more experience.

So, the moment you define the problem in this way, it clearly indicates that you need to select a tool from diagnostic statistics, right. So, here you need to, you can build a plot right, you can make a graph who is leaving more, experienced people or inexperienced people or people who are having specific skill, those people, so which type of people are leaving more. So, that is what you, that is what, so in the first step that what I am trying to make you understand, first thing define your problem clearly and why it is important because of two reasons. First is what type of data you need to collect that is dependent on your definition of your problem, and second, which type of tool you will use to analyze the data that is also dependent. So, if you are not able to define this problem or ill-conceived it is not appropriate.

In that case, you may not be able to achieve the desired goal which you are expecting to achieve it. So, this is the first step that is why it is always important to define your problem clearly. Whatever problem that you are dealing with is related to HR because we are discussing HR, so you can take functions and then relate to any HR; any HR-related function defines what the problem is and then think about the data and tool; it is dependent on that. So, if your problem is not defined clearly then you may face this problem. So, this is the first step of the data science framework.

Let us move to the second. So, now question comes what a HR problem should cover? So, a statement of what a statement of why and a statement of the desired outcome. So, these are the three things that a problem statement should cover, right? So, statement of what? So, what is the problem? Attrition is the problem. Why attrition is a problem, right? Because people are less paid, people are more experienced, culture right, department, supervisor. So, why that problem exists? I hope you can understand the attrition that we are talking about. So, why people are leaving? So, through the data that you have to explain, right? And now what exactly that you want to do it or what you are trying to achieve it? So, you want to achieve the percentage of people who are leaving and you want to know in which department most of the people are leaving.

So, that is what you want. In the example that you can take related to the HR operations efficiency, so in how many days you are able to complete the recruitment, what is the cost of recruitment per application right that per-application cost, per candidate selected candidate cost right what is the cost? So, that these are the matrix that you can calculate and then you can divide whether HR operations were efficient or not. But you have clearly identified for an efficiency these are the parameters that you are, that you want to calculate, right? So that is what you have to define. So basically it is asking about the matrix that is what on through which you want to analyze that particular phenomena. So these all three things your problem statement should have in order to make a good problem statement related to the HR analytics. So, statement of what, statement of why and the matrix which through which you will analyze this particular phenomena.

If you have these three things you will not face any problem. So, I have given you the example of HR efficiency. So, I have taken the example of this recruitment. So, in the recruitment. So, what is the problem? Number of applications are less or more related to, why it is less? Because of the advertisement.

So, now you can calculate how much you have invested per application. So, let us assume you receive 10,000, so what is the amount and the total amount that you can divide by the 10,000, the total amount that you have invested for the job advertisement. So that is how you can calculate per application cost that you have invested. So that matrix that you have already with you, through this you want to describe this particular problem. So these are the three things that you should have when you are defining the HR problem.

So I hope you would have understood. Next is collect the data and manage the data, right? So, from where you will collect the data most of the organizations are having HRIMS. So, this could be the good source of data, right. Sometime when you will develop the matrix that time this HRIMS may not provide the complete data. This is required to analyze the data, which is required to analyze data that may not be, may not provide you with the complete data because some data is available in the marketing department, some data is available in the finance department, and some data is available in the operations department. So, sometime this HRIMS may not provide you the complete information, then what do you have to do? You have already developed the matrix then think what could be the data source for the remaining data.

So, this is the first step. So, you have checked your HRIMS and which data is not available think about that and then you can collect that data also. So, the moment you have collected the required data from various sources then what you can do? Then you can define what are the dependent variable and independent variable, right. So, dependent

variable is interest variable that is what I would say. So, for example, this work experience, right. If you want to do study related to the work experience, right and second thing salary dissatisfaction.

So, such kind of variables that may come under the dependent variable. So, simple I would say. Which variable is your interest variable. Second is independent variable, what is the reason why more experienced people are applying to your organization, right. So, work experience is your study variable, dependent variable, interest variable, why more work experience people are applying, that is what you want to know.

So, whatever is the reason you can say that because of employer branding, because of the salary that you provide the kind of culture that you have. So, these culture, salary, location, these all things will come under the independent variable because they because of this they more experienced people are applying to your organization. So, that is the that is how you have to define after the collecting data. How you will decide this dependent variable? Which one is a dependent variable? You can analyze what? You can analyze your problem statement. So, in a problem statement what is your interest, what exactly that you want to know.

So, from there, you can drive, or you can reach the dependent variable and then from there, whatever data is available, what could be the possible cause of that particular dependent variable, and why it is happening? So, through the independent variable that you can do. So, when you are trying to establish some relationship between independent and dependent variable, you should have the logic behind it. So, that LAMP model if you remember in the first session that is what we had discussed the LAMP model. So, without any logic, please do not consider all variables as an independent variable, right. So, you should have the logic, this thing may lead to this particular thing, right.

If that is not there, in that case you need to think about it, you need to think. So, all independent variable that you have to identify through the logic. If logic is not there, in that case you need to be worry about it. Without logic, do not select the independent variables. There has to be a some logic, then only you apply or you use that variable as a independent variable.

So, the moment you have defined this variable, independent variable, dependent variable, whatever data was not there that also you have collected in a excel format. So, now you need to perform three things, extract, transform and load, extract, transform and load. So, data that you have extracted, if that data is not as per the requirement, requirement for what? For the analysis. For example, you need the mean value correctly, but the mean value is not there correctly. Then what you can do you can calculate the mean value and

transform                      that                      particular                      data                      right.

So, and then you can load on a software through which you have to analyze the data. So, software the data visualization tool that we will learn in this course pivot table, through excel, power bi and tableau. These are the three tools that we are going to learn. So, before analysing the data through these three tools, we need to transform that particular data and then you have, then we have to load. So, these three terms that you can remember,                      extract,                      transform                      and                      load.

So, whichever data is not available then you have to extract. If it is not up to the mark, it means it is not in a same format in which we require to analyze the data then we need to transform and then we need to load. So, these things that we will learn when we will use the software in this course. So, I hope you would have understood and second thing last and not last I would say most important thing related to this data that we need to understand the trustworthiness. If data is not trustworthy, data should be trustworthy, if it is not trustworthy otherwise garbage in and garbage out, whatever you will put you will get out and you will not be able to make the meaningful results.

So that is why you need to be very very careful about your data. You should check the trustworthiness of that particular data also. So, I hope you would have understood how to collect the data and manage the data. Next, build the model. So, that model that I was talking about, so that matrix that you had decided to, matrix describe the problem to understand the problem to find out the solution, right.

So, now you have to decide the input data. So, if you remember that dependent variable, independent variable, right is descriptive. So, all data that you have collected in excel format, so that will consider as a input data. So, the moment you have identified this input data. Now you need to process this particular step. So now in a process step that you already know four type of analytics that we had discussed.

So descriptive, diagnostic, predictive and prescriptive. So in this step after finalizing the input data you need to apply this particular analytical tool to analyze that data and then you will get the output. So if this data is trustworthy your output also will be trustworthy. If data is not trustworthy, then your output also cannot be trusted and you cannot take the decision based on this. So, that is how you need to build the model, you need to input and you need to check the output and you can apply.

So, this is in three format like input, process and output. So, input data that you have already identified, based on your problem statement, you have already finalized the tool. So, after finalizing the input data, now you need to apply the analytical tool, it could be

descriptive, diagnostic, predictive, prescriptive and then you will get the statistical outcome of it, right. Then you will get a statistical outcome of it, right. Next, so whatever model that you have developed, now you have to evaluate and critique the model. So, in this step what exactly that you are supposed to do, in this step you have to evaluate.

So, whatever variable that you have collected, why it is you have collected, how close it is to the problem that is what you need to understand. Because every matrix that you through which you collect the data that does not describe the exact problem of the related to your goal. But each matrix that you are collecting should have a higher level of proximity to describe that particular problem or to identify the solution. So, if that proximity is not to the highest level, right, for example, you are having two matrix, right, both matrix are describing the problem, but one is close and another one is not that close, right. So, for example, that I talk about the effectiveness of the recruitment and selection activity, right.

So, two numbers are there, number of application that is the one of the matrix and second one is quality of hire, right. So, both on the basis of both matrix you can establish the effectiveness of the recruitment right. So, one effectiveness that you can say number of application that has increased as compared to the last year and based on that you can say that whether this recruitment was effective or not. Second that quality of people who have applied to the available position based on that also you can say whether the recruitment was effective or not. But what we want? We want quality of people or number of applications, whether they are appropriate or not.

So, it is obvious we want quality of people, right, both. So, in this case you can see this quality of hire is having more proximity to the effectiveness of recruitment than the number of application. Although both matrices are very very important to understand the effectiveness of the recruitment but as far as my concern this quality of hire is more appropriate than the number of application. So, that is why I am saying so you might be having multiple matrices related to one particular problem. So which one you have to select, which one is more closer to the problem, right. So that is how you have to evaluate and critique that particular model.

So that is how you have to critique that particular problem, particular model, right and after critic after this then you have to present the result whatever results that you are getting it right and if you find these results are suitable and then you can deploy that particular model right. So, thank you. I hope you would have learnt the data science framework in this session. Thank you.